to wonder whether poking the field in one box might be able to cause drastic changes in the rest of the state, even very far away.

Indeed it can. By poking a quantum field in one tiny region of space, it's possible to turn the quantum state of the whole universe into *literally any state at all*. Technically this result is known as the *Reeh-Schlieder theorem*, but it has also been called the *Taj Mahal theorem*. That's because it implies that without leaving my room, I can do an experiment and get an outcome that implies there is now, suddenly, a copy of the Taj Mahal on the moon. (Or any other building, at any other location in the universe.)

Don't get too excited. We can't purposefully force the Taj Mahal to be created, or reliably bring anything particular into existence. In the EPR example Alice can measure her spin, but she can't guarantee what outcome that measurement is going to get. The Reeh-Schlieder theorem implies that if we measure quantum fields locally, there is some measurement outcome we could get that would be associated with a Taj Mahal suddenly being on the moon. But no matter how hard we try, the probability of actually getting that outcome will be really, really, really tiny. Almost all the time, a local measurement leaves distant parts of the world pretty much unaltered. Like many remarkable results in quantum mechanics, it's not a practical worry.

A popular after-dinner discussion among certain circles is "Should we be surprised by the Reeh-Schlieder theorem, or not?" It certainly seems surprising that we can do a measurement in our basement that turns the state of the universe into literally anything. As surprising things go, that's up there. But the other side argues that once you understand entanglement, and appreciate that things can technically be possible but are so incredibly improbable that it really doesn't matter, we shouldn't be very surprised after all. Looked at in the right way, the potential for a Taj Mahal on the moon was there all along, in some tiny part of the quantum state. Our experiment simply lifted it out of the vacuum by branching the wave function in an appropriate way.

I think it's okay to be surprised. But more important, we should appreciate the richness and complexity of the vacuum. In quantum field theory, even empty space is an exciting place to be.